



Pushing the limits of performance: Supercomputing on Azure AI Infrastructure

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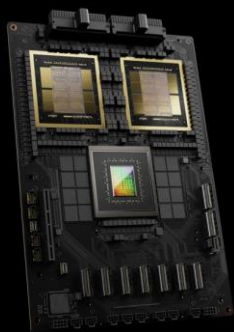


Supercomputing Operations Overview

ND v6 GB200 and GB300

GB200 and GB300 Superchip Module

1x Grace CPU, 2x Blackwell GPUs



VM

GB200 & GB300 Compute Tray

(VM and Node)



NCCL+
NVLink

GB200 & GB300 NV72

72 liquid cooled, NVLINK connected GPUs per rack



NCCL
+
NDR

InfiniBand Networking powering DC Scale Supercomputing

Thousands of ND v6 GB200/GB300 VMs



Grace Blackwell Superchip

2x B200 or B300 Blackwell GPUs

- 1 Grace CPU with ARM Neoverse v2 cores
- B200: 192GB of HBM3/GPU
- B300: 288GB of HBM3/GPU
- New FP4 datatype, with 1.5x higher FP4 in B300

Compute Tray

- 2x GB Superchips
 - 2 Grace CPUs
 - 4 Blackwell GPUs
- Next Gen Azure Boost FE NIC

NVL72 Exa-scale Rack

- 1.8TB/s 5th Gen NVLINK connecting 72 GPUs in rack
- Massive pool of shared HBM
 - 14TB on GB200
 - 20TB on GB300
- 1,800 GB/s B/W per GPU all-to-all and all-reduce in rack

InfiniBand Scale-out

- GB200: 400Gb/s IB
- GB300: 800Gb/s IB
- Non-blocking fat-tree network
- Scale out to 100Ks of GPUs

Architecture Overview

High-Density GPU Configuration

Each rack integrates 36 dual-GB200 nodes, combining for a total of 72 Hopper GPUs interconnected through the NVLink Switch System for extreme AI performance.

NVLink Interconnect Fabric

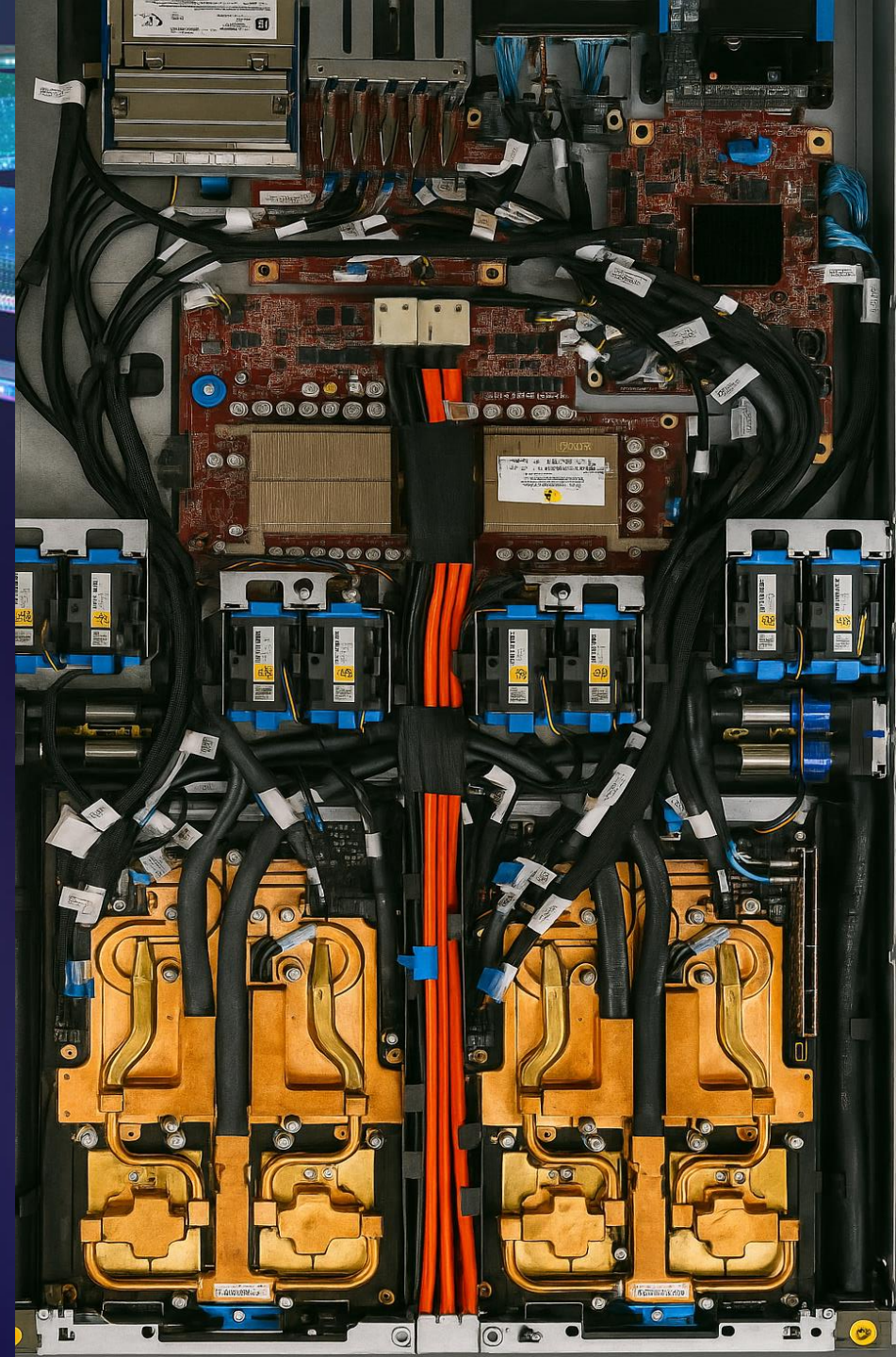
NVLink fabric delivers 7.2 TB/s bandwidth, enabling fast, unified memory access and communication across GPUs.

Scalable NVSwitch Domains

NVSwitch domains enable scalable deployment for training trillion-parameter AI models with high computational power.

Efficient Cooling System

Liquid cooling maintains optimal GPU temperatures for reliable and efficient high-performance operation.



Supercomputing Operations Overview

Key Operational Strategies

High-Speed GPU Communication

Utilize NVLink and NVSwitch to enable high-bandwidth, low-latency communication between GPUs for efficient data processing.

Telemetry and Anomaly Detection

Monitor GPU health and performance continuously using telemetry systems to detect anomalies early and prevent failures.



72 GPUs (18 nodes) in a Single NVLINK connected Exa-scale Computer
Massively Scalable AI Supercomputer up to 100Ks of GPUs

Interconnect & Scalability

NVSwitch Fabric Communication

NVSwitch fabric enables full GPU-to-GPU communication within a rack for high throughput and low latency.

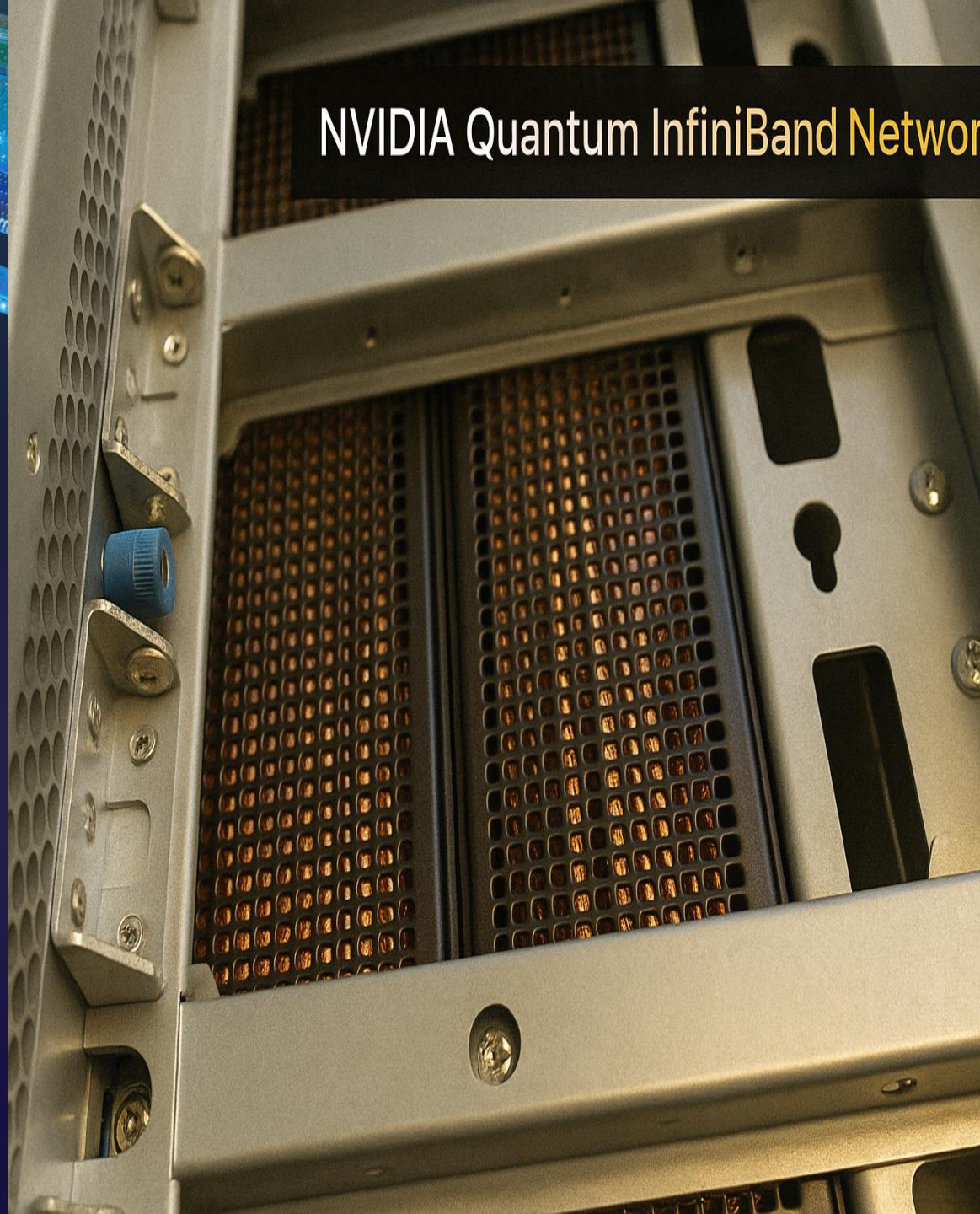
InfiniBand Quantum-2 Connectivity

InfiniBand Quantum-2 provides low-latency, high-bandwidth connections across racks for distributed training.

NVLink Switch System Scaling

NVLink Switch System manages data flow between GPUs and nodes enabling scalable deployments.

NVIDIA Quantum InfiniBand Network



Node Recovery Lifecycle

Anomaly Detection

Uses GPU telemetry and XID (149) events to identify potential system issues early.

Automated Recovery Processes

Triggers BMC resets and node isolation to restore system functionality automatically. node isolation through automation to minimize system downtime.

Validation and Health Checks

Performs performance tests and health checks to ensure node is fully operational.

Incident Logging and Monitoring

Integrates with Azure and telemetry systems for detailed incident records and analysis.



Context - Why Cluster Validation Matters

- Microsoft builds and maintains GPU superclusters optimized for large-scale foundation model training.
- These systems comprise thousands of interconnected GPUs across multiple racks and nodes, linked through NVSwitch, NVLink, and InfiniBand fabrics.
- At this scale, even a single underperforming node for instance, one with degraded NVLink bandwidth or a throttled GPU can impact global training efficiency and convergence stability.

Motivation for Validation

- The objective is to guarantee consistent, high-throughput performance across every GPU and node before large-scale training begins.
- Validation detects and isolates issues in:
 - Compute path
 - Communication path
- This ensures a homogeneous, predictable cluster environment that minimizes scaling inefficiencies and simplifies root-cause analysis during large-scale runs.

Health Check vs. Performance Validation

- Health Check: Verifies that all system components are operational and free of critical faults.
 - Focuses on **hardware integrity and stability**.
- Performance: Evaluates compute and communication efficiency relative to known baselines.
 - Focuses on **throughput, latency, and consistency** across GPUs/nodes.

Performance Benchmarks

- **GEMM micro-benchmarks (cuBLASLt)** — measure per-GPU TFLOPS.
- **NCCL collectives (AllReduce, AllGather)** — validate communication bandwidth and topology health.
- Small scale training run
- Large scale training run
- Outcome: node or cluster *meets or exceeds expected performance variance thresholds* (e.g., $\pm 3\text{--}5\%$ of mean).

Microbenchmarks - GEMM

m	n	k	ND GB200 v6 (TFLOPS)
1024	1024	1024	349.4
2048	2048	2048	1676.1
4096	4096	4096	2694.7
8192	8192	8192	2781.5
16384	16384	16384	2819.5
32768	32768	32768	2755.3
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1024	2145	1024	627.2
6144	12288	12288	2798.7
802816	192	768	1482.5

Microbenchmarks - NCCL

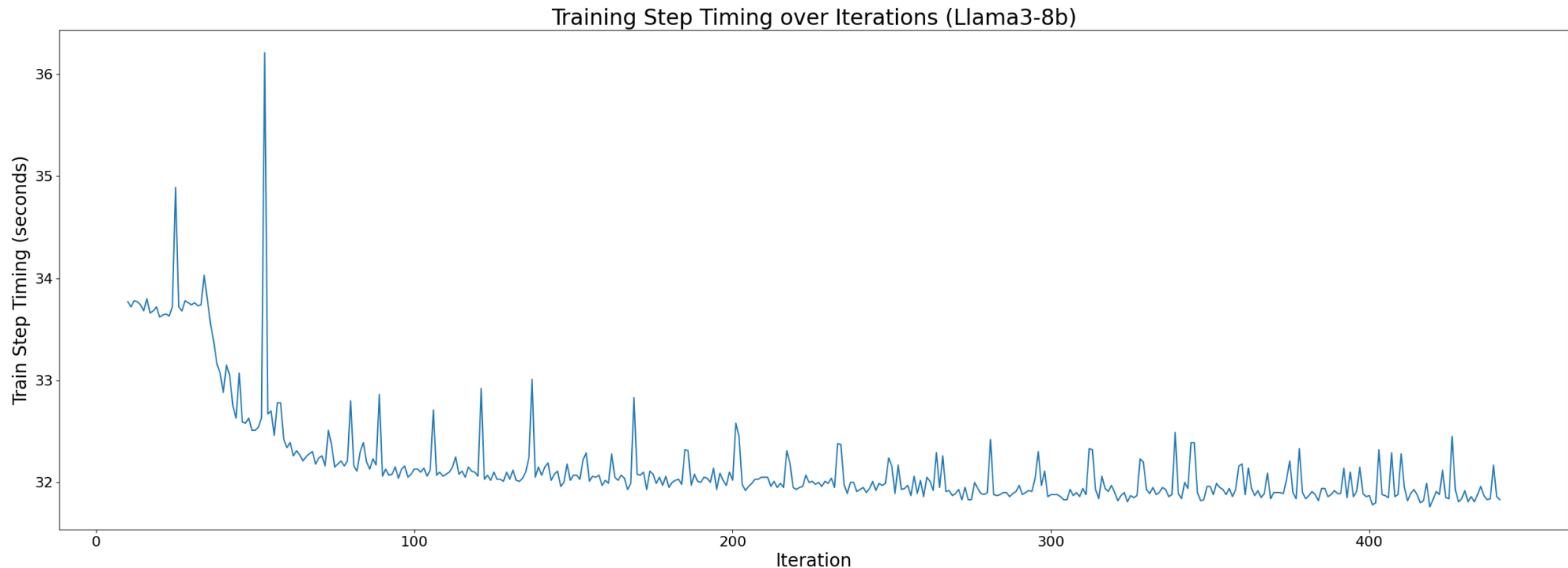
- Why we run NCCL tests
 - Verify that NVLink (within node) and network fabric (across nodes) are healthy.
 - Establish *communication ceilings* expected during real training.
- Within-node GPU-to-GPU AllReduce
 - Tests the local topology: NVLink / NVSwitch.
 - Expected to hit high bandwidth with low variance.
 - If this is slow → training will bottleneck even before leaving the node
- Node-to-node: Multi-node AllReduce
 - Measures performance over IB/RoCE/Ethernet
 - Shows algorithm BW vs bus BW scaling across message sizes.
 - If this is slow → training steps stretch due to gradient sync delays.

Training vs. Inference for Cluster Validation

- Training workloads:
 - Exercise forward + backward passes, optimizer states, and collective communication across GPUs.
 - Reveal bottlenecks like NCCL imbalance, power throttling, or memory bandwidth issues.
- Inference:
 - Exercises only the forward path and minimal communication
 - Useful for kernel-level testing.
- Hence: Training workloads provide a comprehensive, realistic view of cluster performance and stability.

Small scale training workloads

Small-scale training enables precise identification of performance or hardware bottlenecks of single nodes ahead of large-scale workloads.



Large scale training

- Extends validation from single-node to multi-node workloads (2 + nodes, either within a rack or across racks).
- Captures communication efficiency, scaling behavior, and synchronization stability under distributed training.
- Monitors average step time, throughput per GPU, and variance across nodes to detect scaling inefficiencies.
- Confirms that compute and communication balance holds as we move beyond a single node.

Grok1 Workload

- Grok-1 is a Mixture-of-Experts (MoE) transformer, making it a strong representative for modern large-scale distributed workloads.
- Activates only a subset of experts per token, creating dynamic and uneven communication patterns across GPUs.
- Stresses both compute and communication simultaneously.
- Helps evaluate scaling efficiency, load balance, and inter-node bandwidth utilization under realistic training conditions.

Training Results



Training Results

